

# Pythia-14M Repeat-Match Checkpoint Trajectory

Attention-role probes versus causal head ablation

Branch specialization research project

2026-05-22

Checkpoint reason: this is the first real-transformer test of the probe-before-causality pattern found in the routed toy model.

# Question

The routed toy model showed:

*gate alignment can appear before causal branch modularity.*

Pythia has no router gate, so the real-model analogue is:

*does an attention-role probe appear before strong causal head importance?*

We test this on repeat-match heads across Pythia-14M training checkpoints.

- Models: EleutherAI/pythia-14m-seed1, seed2, seed3
- Revisions: step0, 64, 256, 1000, 4000, 16000, 64000, 143000
- Probe: attention from second occurrence of  $x_i$  to first occurrence of  $x_i$  in repeated-token sequences
- Causal test: zero selected head outputs and measure second-half next-token loss
- Heads tested: top repeat-match head in layers 0 and 1
- Controls: 8 random same-layer head sets per seed and checkpoint

Probe selection and causal evaluation use separate synthetic sequence sets.

# Metric Glossary

**Probe specialization** Normalized repeat-match attention score of the selected heads. This measures whether the head has the attention pattern.

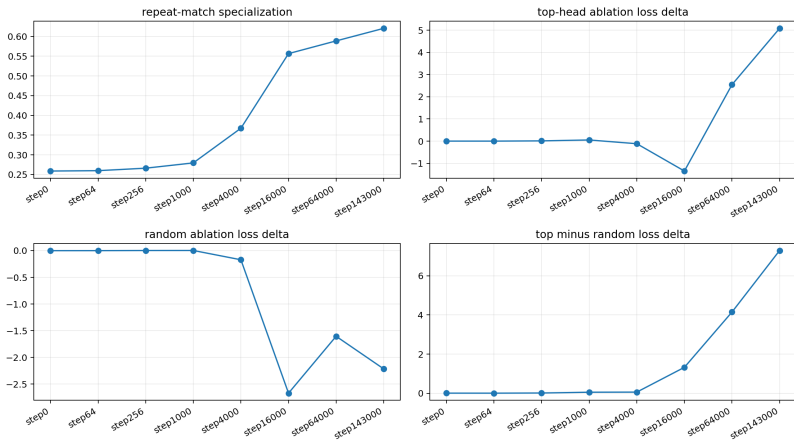
**Top loss delta** Loss after ablating selected repeat-match heads minus baseline loss.

**Random loss delta** Same ablation metric for random same-layer heads.

**Top minus random** Top-head loss delta minus random-head loss delta. This is the primary causal metric here.

The relative metric matters because some random head ablations improve this synthetic loss at intermediate checkpoints.

# Trajectory



Probe specialization starts rising before the top-minus-random causal excess becomes large. The causal excess is the bottom-right panel.

# Aggregate Results

| Revision   | Probe spec. | Top loss | Random loss | Top – random |
|------------|-------------|----------|-------------|--------------|
| step0      | 0.2590      | 0.0018   | -0.0007     | 0.0025       |
| step1000   | 0.2796      | 0.0527   | 0.0027      | 0.0500       |
| step4000   | 0.3675      | -0.1155  | -0.1711     | 0.0556       |
| step16000  | 0.5566      | -1.3471  | -2.6695     | 1.3224       |
| step64000  | 0.5888      | 2.5466   | -1.6040     | 4.1506       |
| step143000 | 0.6206      | 5.0833   | -2.2149     | 7.2982       |

Means over 3 seeds. The probe rises by step4000, while causal excess remains small until later checkpoints.

## Supported

- Repeat-match attention specialization can precede strong causal importance.
- This mirrors the toy result at the measurement level: probe/gate signals are not enough by themselves.

## Not shown

- This does not show branch-level modularity in Pythia.
- These are ordinary attention heads, not routed branches.

- Pythia-14M pilot only; Pythia-160M remains the stronger target.
- 3 seeds only.
- Layers 0 and 1 only.
- Synthetic repeated-token task.
- Head-output ablation, not full path patching.
- Some random ablations improve synthetic loss, so use relative excess over random controls as the main causal readout.



# Project-Level Update

The project now has the same methodological lesson in two settings:

- ① Routed toy model: gate alignment preceded causal branch modularity.
- ② Pythia-14M: repeat-match specialization preceded strong causal head importance.

The defensible paper framing is:

*specialization probes must be paired with causal tests, because the two can have different developmental timelines.*

# Decision And Next Step

**Decision.** Keep the probe-versus-causality trajectory as a central measurement axis.

**Next step.** Run the same checkpoint-trajectory analysis on Pythia-160M:

- use the already validated repeat-match layers;
- add cross-seed raw-score alignment;
- compare aligned transferred heads against same-index transferred heads across checkpoints.

## Source

- `scripts/pythia_repeat_match_checkpoint_trajectory.py`
- `doc/phase1_pythia14m_repeat_match_checkpoint_trajectory.md`

## Copied data

- `data/revision_summary.csv`
- `data/revision_seed_summary.csv`
- `data/probe_head_scores.csv`
- `data/ablation_results.csv`
- `figures/repeat_match_checkpoint_trajectory.png`

## Run command

- `python3 -u scripts/pythia_repeat_match_checkpoint_trajectory.py --model-size 14m --seeds 1 2 3 ...`